

6th Grade Math Review

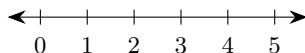
Name: _____ Date: _____

Directions

No calculator. Write decimal answers as decimals. Write fraction answers in simplest terms. Show work in space provided or on scrap paper as directed.

A. Number Lines

1. Plot the point 4 on the number line below:



B. Order Of Operations

- | | |
|--------------------------------|-----------------------------|
| 1. $8 + 3 \times 4 =$ _____ | 4. $(5 + 1)^2 =$ _____ |
| 2. $(21 - 4) \times 3 =$ _____ | 5. $3(2^2) =$ _____ |
| 3. $2^4 + 1 =$ _____ | 6. $(3 \times 2)^2 =$ _____ |

C. Decimals, Fractions, And Percents

- | | |
|---|---------------------------------------|
| 1. In 82.731, the tenths digit is _____ | 5. $65\% =$ _____ <i>as a decimal</i> |
| 2. $314.7 + 29.58 =$ _____ (<i>decimal</i>) | 6. $0.8 =$ _____ <i>as a fraction</i> |
| 3. $81.5 - 7.94 =$ _____ (<i>decimal</i>) | 7. 15% of \$80 = _____ |
| 4. $0.32 =$ _____% | |

D. Facts And Decimal Multiply/Divide

- | | |
|-------------------------|--|
| 1. $9 \times 6 =$ _____ | 3. $2.4 \times 1.5 =$ _____ (<i>decimal</i>) |
| 2. $63 \div 9 =$ _____ | 4. $5.6 \div 0.8 =$ _____ (<i>decimal</i>) |

E. Ratios, Fractions, And Proportions

- | | |
|---|---|
| 1. Greater: $\frac{5}{8}$ or 0.61? _____ | 3. 4 snacks cost \$20. 7 snacks cost _____ dollars. |
| 2. $\frac{4}{7} = \frac{?}{21}$? = _____ | 4. $\frac{5}{6} = \frac{x}{18}$ $x =$ _____ |

F. Vocabulary, Expressions, And Equations

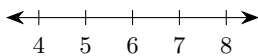
- | | |
|---------------------------------------|--|
| 1. In $7m$, 7 is the _____ | 4. Undo $\times 5$, use inverse op: _____ |
| 2. In $n + 8 = 12$, n is the _____ | 5. Combine like terms: $6y - 2y =$ _____ |
| 3. Undo $+8$, use inverse op: _____ | 6. If $n = 4$, then $5n - 3 =$ _____ |

G. Distributive Property

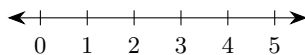
1. Expand: $3(y + 6) =$ _____

H. Equations, Inequalities, And Graphing

- | | |
|---|-----------------------------|
| 1. $x + 7 = 18$ $x =$ _____ | 3. Solve: $x + 3 > 8$ _____ |
| 2. $\frac{x}{4} = 6$ $x =$ _____ | 4. Solve: $x - 4 < 2$ _____ |
| 5. Write an equivalent way to express $6 < y$: _____ | |
| 6. Graph $6 < y$ on a number line: | |



7. Graph $x > 2$ on a number line:



I. Geometry

- Rectangle with length 9 and width 4: Area = _____
- Rectangle with length 11 and width 3: Perimeter = _____
- Rectangle with area 42 sq units and width 6: Length = _____

J. Units And Quantities

1. 3 hours = _____ minutes
2. $\frac{3}{5}$ of 35 = _____
3. 210 miles/3 hours = _____
4. 72 inches = _____ yards
5. 150 miles/5 gallons = _____

K. Expressions, Factors, And Fraction Structure

1. Is $8x + 1$ a linear expression? _____
2. Prime factorization of 90 = _____
3. Find the GCF of 27 and 36: _____
4. Simplify $\frac{27}{36} =$ _____
5. Convert $2\frac{2}{3}$ to an improper fraction: _____
6. Which is greater: $1\frac{3}{4}$ or $(1)(\frac{3}{4})$? _____
7. In $\frac{m}{2}$, coeff of m is _____
8. In $\frac{q}{5}$, coeff of q is _____
9. Commutative: $8 + 5 =$ _____
10. Commutative: $7 \times a =$ _____
11. Associative: $(b + 3) + 9 =$ _____

L. Distributive Property And Factoring

1. Factor: $8x + 16 =$ _____

M. Decimal Computation

1. $428.5 + 37.26 =$ _____ (*decimal*)
2. $700.32 - 84.67 =$ _____ (*decimal*)
3. $18.4 \times 12 =$ _____ (*decimal*)
4. $168.8 \div 8 =$ _____ (*decimal*)
5. $4.2 \times 1.5 =$ _____ (*decimal*)
6. $7.2 \div 0.8 =$ _____ (*decimal*)

N. Fraction Operations And Properties

1. $\frac{1}{2} + \frac{1}{4} =$ _____ (*simplest*)
2. $\frac{5}{6} - \frac{1}{3} =$ _____ (*simplest*)
3. $\frac{5}{8} \times \frac{4}{15} =$ _____ (*simplest*)
4. $\frac{7}{8} \div \frac{1}{4} =$ _____ (*simplest*)
5. Commutative: $7 \times a =$ _____
6. Associative: $(x + 5) + 8 =$ _____
7. Is subtraction associative? Compare $(12 - 4) - 3$ and $12 - (4 - 3)$: _____